

FACTSHEET FS18 · The portfolio I would actually run

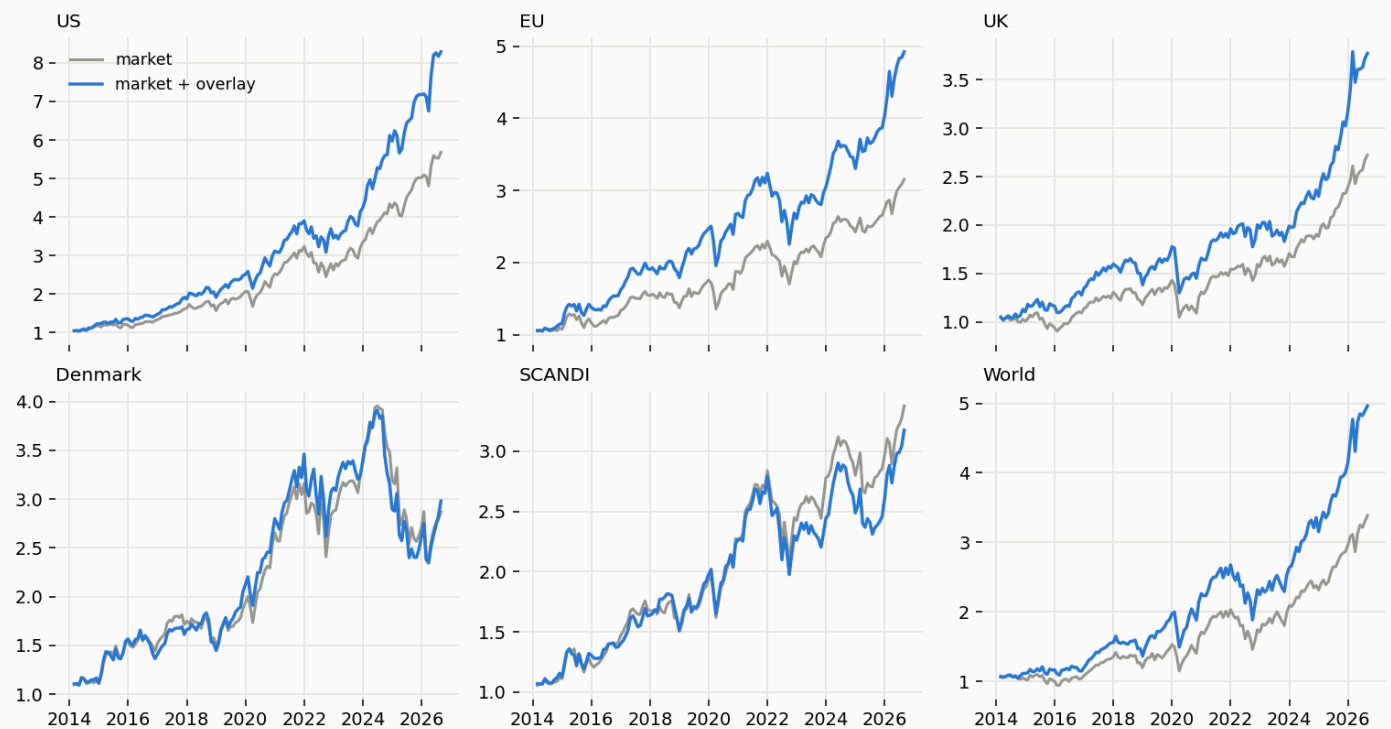
Everything that survived the series, net of realistic costs, next to a plain market portfolio

Henrik Gjerner · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · February 2014 – 2026

Key facts	
Pre-registration	planning/PREREG_FS18.md, saved to the Factsheets folder before any FS18 number was computed; no deviations
Portfolio	100% size-weighted market (long-only) + an overlay scaled to 5% annual volatility
Overlay books	low beta and 12-1 momentum (all markets); plus profit growth and debt issuance composites (US only, FS13b); all beta-neutral with the turnover buffer
Costs	FS16 model: liquidity-dependent spread + square-root impact ($Y = 0.7$) at \$50m per book , plus size-tiered borrow and financing; sensitivity at \$250m
Weights	inverse 36-month volatility per book (equal weights before 36 months); overlay scaled by its trailing 12-month volatility, cap 3×
First month	February 2014: the 12-month volatility estimate needs a year of overlay history
Primary test	mean monthly overlay return pooled over US, EU, UK, Denmark and SCANDI: Newey–West $t > 2$ and positive in 2013–19 and 2020–26

In one paragraph. Not passed, narrowly. Pooled over the five markets the overlay adds **+1.9% a year (t 1.85)**, positive in both halves (+3.0% in 2014–19, +0.9% in 2020–26) but short of the pre-registered $t > 2$. Market by market the picture splits cleanly. In the **US, EU, UK and World** the overlay has a Sharpe ratio near 0.5–0.6, is uncorrelated with the market and lifts the portfolio's Sharpe ratio by 0.15–0.21 at \$50m per book. In **Denmark and SCANDI** it adds nothing, as FS13 and FS16 predicted. At **\$250m per book** the gain shrinks sharply outside the US and turns negative in the UK, Denmark and SCANDI (pooled +0.2%, t 0.2). The honest verdict: a small, cheap-to-run overlay in large, liquid markets is worth having; it is not a strategy that scales, and it is not a reason to trade the Nordics.

Growth of \$1, 2014–2026: market alone vs market + 5%-volatility overlay, net of costs at \$50m per book



1. Market by market

Market	Books	Overlay return (t)	Overlay Sharpe	Correlation with market	Market Sharpe	Market + overlay Sharpe	Gain at \$50m	Gain at \$250m
US	low beta, 12-1 momentum, profit growth, debt issuance	+3.1% (t +2.6)	0.55	-0.15	0.89	1.08	+0.19	+0.17
EU	low beta, 12-1 momentum	+3.7% (t +2.5)	0.60	-0.11	0.58	0.79	+0.22	+0.09
UK	low beta, 12-1 momentum	+2.8% (t +1.8)	0.46	-0.04	0.54	0.69	+0.15	-0.09
Denmark	low beta, 12-1 momentum	+0.5% (t +0.3)	0.08	+0.01	0.47	0.47	+0.00	-0.09
SCANDI	low beta, 12-1 momentum	-0.4% (t -0.3)	-0.08	-0.09	0.62	0.57	-0.05	-0.13
World	low beta, 12-1 momentum	+3.3% (t +2.4)	0.57	+0.03	0.59	0.75	+0.16	+0.11

Sharpe ratios of excess returns over the US T-bill rate (French data); market in USD for World, local currency elsewhere; overlay returns are self-financing and net of every cost in the model. Gain = Sharpe of market + overlay minus Sharpe of the market alone.

2. Drawdowns and sizing

Market	Market max drawdown	Market + overlay max drawdown	Overlay max drawdown	Average overlay scale
US	-24%	-21%	-7%	0.75
EU	-26%	-30%	-13%	0.48
UK	-27%	-27%	-13%	0.48
Denmark	-40%	-40%	-19%	0.49
SCANDI	-27%	-29%	-22%	0.64
World	-28%	-30%	-7%	0.50

The overlay reduces the worst drawdown only in the US (by about 3 points); elsewhere it leaves it unchanged or slightly deeper. Its value is return per unit of risk, not protection. The average scale well below 1 means the raw book mix ran above 5% volatility; at 5% target volatility the overlay is small next to a market leg with 15–20% volatility.

3. The books inside the overlay (\$50m per book)

Market	Book	Net return a year	Sharpe	Monthly turnover	Weight, last month
US	low beta	+2.8%	0.11	15%	0.11
US	12-1 momentum	+3.4%	0.24	29%	0.22
US	profit growth	+4.9%	0.44	28%	0.31
US	debt issuance	+1.0%	0.12	14%	0.36
EU	low beta	+5.8%	0.30	18%	0.38
EU	12-1 momentum	+4.3%	0.30	28%	0.62
UK	low beta	+8.9%	0.45	20%	0.40
UK	12-1 momentum	+2.4%	0.16	28%	0.60
Denmark	low beta	+1.7%	0.11	11%	0.47
Denmark	12-1 momentum	-0.4%	-0.03	28%	0.53
SCANDI	low beta	-0.4%	-0.03	11%	0.52
SCANDI	12-1 momentum	-1.1%	-0.10	25%	0.48
World	low beta	+4.8%	0.23	18%	0.33
World	12-1 momentum	+5.2%	0.41	27%	0.67

4. What the whole series leaves standing

Question	Answer	Where
Do the famous trading rules work as sold?	Mostly not after costs, borrow and multiple testing	FS01–FS12, FS00
Which price signals survive across markets?	Low beta (beta-neutral) and 12-1 momentum are the most consistent; none survives the correction on its own	FS13, FS13c
Do fundamentals add anything?	Yes: profit growth and issuance, in the US stock by stock and outside the US at factor level	FS13b, FS17
Does a clever multifactor selection rule beat equal weights?	No, in 2013–26 or in 1999–2013	FS14, FS15
Does the shortlist hold up before 2013?	Yes, out of sample (Sharpe 0.8, 1999–2013)	FS15
How big can it get?	US about \$1bn; Europe and the UK tens of millions; the Nordics never	FS16
Does it improve a market portfolio?	A little, in large markets, at small size; not significantly when pooled	FS18

5. Caveats

- Not out of sample for the US fundamental legs, which were chosen from 2013–26 results (FS13b). FS15 is the out-of-sample evidence for the price legs.
- Cost model parameters (spread law, $Y = 0.7$) are literature values, not calibrated to our own fills.
- The overlay is scaled on its own trailing volatility; a sharp volatility regime change can leave it over- or under-sized for months.
- Market portfolios are our point-in-time universes, not investable indices; no fees are charged on the market leg.
- FS18 adds 14 trials to the ledger (now 638).

6. References

- Frazzini, A. & Pedersen, L. H. (2014). Betting against beta. *Journal of Financial Economics* 111(1), 1–25.
- Jegadeesh, N. & Titman, S. (1993). Returns to buying winners and selling losers. *Journal of Finance* 48(1), 65–91.
- Moreira, A. & Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance* 72(4), 1611–1644.
- Frazzini, A., Israel, R. & Moskowitz, T. J. (2018). Trading costs. SSRN working paper 3229719.
- DeMiguel, V., Garlappi, L. & Uppal, R. (2009). Optimal versus naive diversification. *Review of Financial Studies* 22(5), 1915–1953.

7. Reproduce

planning/PREREG_FS18.md → code/fs18.py (→ results/fs18.json) → code/build_fs18.py. Reuses impact.py (FS16) and fundamentals.py (FS13b).